

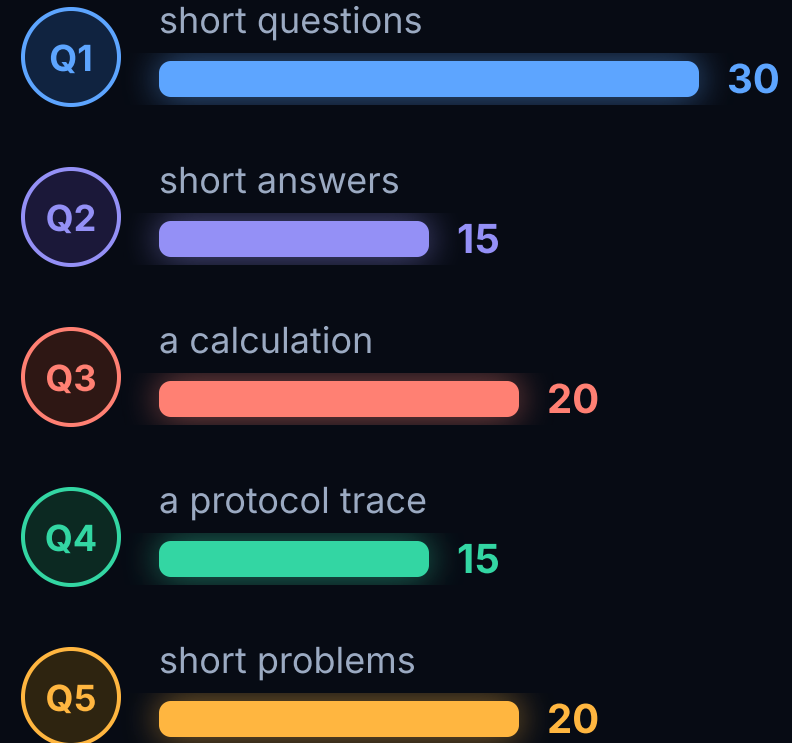
Midterm Review

What it covers, what it looks like

Thursday, October 15, 2:00–3:20 PM. This lecture lists what the midterm covers, shows the five kinds of question on the paper, and works through practice questions in the same form.

Yaxiong Xie

MIDTERM · 100 POINTS



five questions, 80 minutes

When is the midterm, and what can you bring?



Oct 15

Thursday
2:00–3:20 PM



80 min

five questions
100 points



closed

closed book,
no notes



calculator

a basic calculator
is allowed

Thursday, October 15, 2:00–3:20 PM. Closed book; bring a basic calculator.

Which lectures does the midterm cover?

Chapter 1

The big picture

The Internet at scale

Layers

Chapter 2

The application layer

Sockets

The Web and HTTP

DNS

Video streaming

Chapter 3

Reliable data transfer

Pipelining

Congestion control, Parts 1–6

Eleven lectures, from the big picture to Part 6 of congestion control.

What does the paper look like?

01

The paper

Five questions, 100 points, 80 minutes. Each kind of question asks for a different kind of answer.

What kinds of question are on the paper?

	QUESTION	WHAT YOU WRITE	POINTS
Q1	Short questions	ten; a number, a word or a circle each	30 points
Q2	Short answers	three; one to three sentences each	15 points
Q3	A calculation	several steps that build on each other	20 points
Q4	A protocol trace	the steps are given; you fill in what happens	15 points
Q5	Short problems	several short parts on one topic	20 points

45 points for concepts in Questions 1–2, 55 for working through 3–5.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What does a short question look like?

ON THE PAPER: A SHORT QUESTION WITH TWO PARTS

A 4-second video segment is encoded at 3 Mbps.

(i) How large is the segment?

(i) 1.5 MB

(ii) How long does it take to download over a steady 2 Mbps connection? Ignore the round-trip time.

(ii) 6 s

(i) (ii)

one blank per part

answer each part on its own

MB · s

the unit is printed

write only the number

yes / no

some parts are circled

circle one of the words given

Ten short questions; each answer is a number, a word, or a circled choice.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What does a short answer need?

ON THE PAPER: A SHORT ANSWER

A campus replaces its Wi-Fi access links with fiber. Does the code of a web browser on campus have to change? Why or why not?

No. The browser only uses the socket interface of the transport layer.

Wi-Fi and fiber differ in the link and physical layers, below that.

The answer first, then the reason.

Three short answers: the answer and its reason, in one to three sentences.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What does a calculation look like?

A CALCULATION: ONE SCENARIO, SEVERAL PARTS

A 4-second video segment is encoded at 3 Mbps and downloaded over a steady 2 Mbps connection.

(a) How large is the segment?

WORKED OUT

$$4 \text{ s} \times 3 \text{ Mbps} = 12 \text{ Mb} = 1.5 \text{ MB}$$

↓ uses (a)

(b) How long does the download take?

WORKED OUT

$$12 \text{ Mb} \div 2 \text{ Mbps} = 6 \text{ s}$$

One scenario, several parts; each part uses the answers before it.

Will the formulas be given?



Printed on the paper

We print the formulas you need wherever we can.



Known by heart

A few basic relations from the lectures are not printed.

We print formulas wherever we can; a few basic ones you should know.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about Chapter 1?

02

Chapter 1

The big picture, the Internet at scale, and layers. Each slide lists the knowledge points you should know, lecture by lecture.

What should you know about the big picture?

- 1 Two views of the Internet**
The simple service applications see,
and the network of networks inside that provides it
- 2 Edge, access and core**
End systems at the edge, the access network that reaches
the first router, and the core between access networks
- 3 Encapsulation**
What each layer adds on the way down (segment, packet, frame),
and the order in which the receiver removes it

Scale shows where the data travels; layers show the work done on it.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about the Internet core?

- 1 How the core scales**
Why direct links do not scale, and what routers, ISPs, IXPs and content networks each add
- 2 Packet switching**
Packets from many users taking turns on one link, and store-and-forward at every router
- 3 Circuit vs packet switching**
What each one reserves or shares, which one suits bursty users, and the risk of sharing

Routers and ISPs let the core scale; packets share its links on demand.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about delay, loss and throughput?

- | | | |
|---|--------------------------|--|
| 1 | Delay at one hop | Transmission, propagation, processing and queueing delay, and what each one depends on |
| 2 | End-to-end delay | How the delays of every hop add up along a path through several routers |
| 3 | Queueing and loss | Why packets queue at a router, how long a packet waits behind the ones ahead, and when packets are dropped |
| 4 | Throughput | The bottleneck link of a connection, and how flows that share one link divide its rate |

End-to-end delay adds up every hop; the slowest link sets the throughput.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about layers?

- | | | |
|---|-------------------------------------|--|
| 1 | Why layers | How a stable boundary lets one layer change without changing the layers above it, and what layering buys and costs |
| 2 | The five layers | What each of the five layers does, and why the Internet stack has five where OSI has seven |
| 3 | Service, interface, protocol | The service, the interface and the protocol of a layer, and how the three differ |
| 4 | Where layers run | The layers on an end host and on a router, and which nodes take part in each layer's protocol |

A stable boundary lets the layer below it change on its own.

1 The paper

2 **Chapter 1**

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about Chapter 2?

03

Chapter 2

The application layer, sockets, the Web and HTTP, DNS, and video streaming.

What should you know about the application layer?

- | | | |
|---|---------------------------------|---|
| 1 | Client-server and P2P | Who must be reachable and who speaks first, and how the cost grows with the number of users |
| 2 | Application protocols | The four elements every protocol defines: type, syntax, semantics and rules, seen in HTTP |
| 3 | TCP or UDP | What each transport promises and what neither promises, and which applications need which |
| 4 | Byte streams and framing | Why TCP keeps no message boundaries, and how an application marks where a message ends |

An application chooses its architecture, its protocol and its transport.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about sockets?

- 1 The socket calls**
What `socket()`, `bind()`, `listen()`, `accept()` and `connect()` do, in what order, and which calls wait
- 2 Listening and connection sockets**
Why `accept()` returns a new socket, and how the kernel finds the socket for each arriving segment
- 3 Serving many clients**
Walking a list of sockets, a thread per connection, or asking the kernel: what each one costs

One listening socket, one connection socket per client, bytes not messages.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about the Web and HTTP?

- | | | |
|---|-------------------------------|---|
| 1 | Objects and messages | What a page is made of, what a URL names, and what HTTP requests and responses contain |
| 2 | Persistent connections | The round trips a page costs with a new TCP connection per object, with one persistent connection, and with six in parallel |
| 3 | State and cookies | Why HTTP is stateless, and how cookies let a site remember a user |
| 4 | Web caching | What a cache does on a hit and on a miss, and how it changes the average time per request |

One request, one response; persistent connections and caches cut the cost.

1 The paper

2 Chapter 1

3 **Chapter 2**

4 Chapter 3

5 Practice

What should you know about DNS?

- | | | |
|---|------------------------------|--|
| 1 | Why names | Why we look up names instead of using IP addresses, and the four fields of a resource record: name, type, value, TTL |
| 2 | Replication and zones | Why one central server does not work, why copying it is not enough, and how zones and delegation fix it |
| 3 | Name resolution | Recursive and iterative queries, referrals, the round trips of one lookup, and why DNS uses UDP |
| 4 | Caching and TTL | What a resolver caches, how that shortens later lookups, and how the TTL trades update speed against query load |

A lookup walks the tree once; the cache and the TTL make later ones short.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about video streaming?

- | | | |
|---|------------------------------|---|
| 1 | Video and compression | What makes video different from other traffic, how I frames, P frames and resolution set its size, and the bitrate ladder |
| 2 | Segments and manifest | Why a video is cut into short segments at several qualities, and what the manifest lists |
| 3 | Buffer and bitrate | How the buffer grows and drains, what happens when it empties, and how a player picks each segment's quality |
| 4 | CDNs | Why segments are served from caches near viewers, and how a player reaches the nearest copy |

The player watches its buffer and its speed, and picks a quality per segment.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about Chapter 3?

04

Chapter 3

Reliable data transfer, pipelining and sliding windows, and congestion control up to the end of Part 6.

What should you know about reliable data transfer?

- 1 Failures and fixes**
Bit flips, loss, delay and reordering, and the mechanism added for each: check field, ACK, sequence number, timer, buffer
- 2 Sequence numbers**
How resends create duplicates, and how sequence numbers let the receiver discard them
- 3 Setting the timeout**
Why the timeout must follow the round-trip time, and how EstimatedRTT and DevRTT set it

Checksum, ACK, sequence number, timer, buffer: one mechanism per defect.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about pipelining?

- | | | |
|---|--------------------|--|
| 1 | Why pipeline | Why stop-and-wait wastes a fast link, and how large a window keeps it busy (the bandwidth–delay product) |
| 2 | The sliding window | What base, next and W mean, and when the sender may send a new packet |
| 3 | Go-Back-N | What the receiver does with an out-of-order packet, what an ACK means, what a timeout resends, and fast retransmit |
| 4 | Selective repeat | What the receiver buffers and acknowledges, what the sender resends, and what one loss costs compared with Go-Back-N |

$W = R \times \text{RTT}$ fills the link; Go-Back-N resends the window, SR one packet.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

What should you know about congestion control?

- | | | |
|---|----------------------------|---|
| 1 | What congestion is | A buffer that fills because more arrives than a link can send, how long packets then wait, and when they are dropped |
| 2 | Why control is hard | Links shared with unseen flows, no shares handed out and no signal from the network, and a right rate that keeps moving |
| 3 | Costs of congestion | Four costs: longer waits, resending dropped packets, resending late ones, and wasted links; goodput; why throughput falls past capacity |
| 4 | cwnd and probing | What rate to aim for, how cwnd sets the sending rate, how a sender probes for it, and which signal classic TCP uses |

rate = cwnd × MTU / RTT; TCP treats a loss as the sign of sending too fast.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

Can you answer questions in the form of the paper?

05

Practice

Five questions in the same form as the paper, none of them on it. Try each one before the answer appears.

How many cables do ten hosts need?

SHORT QUESTION · CHAPTER 1

1. Ten hosts are connected without any router: every pair of hosts gets its own cable.

(i) How many cables are needed in total?

(i) 45 cables

(ii) One router is added, and every host connects to it with one cable. How many cables are needed now?

(ii) 10 cables

ANSWER

(i) $10 \times 9 / 2 = 45$ cables, one for every pair of hosts

(ii) 10 cables, one per host; the router joins them

N hosts need $N(N - 1)/2$ direct links; through one router they need N.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

How much does a five-person video call send?

SHORT QUESTION · CHAPTER 2

2. Five people are in a video call. Each camera produces a 3 Mbps stream, and each person must see the other four.

(i) All streams go through a server, which sends each stream to the other four. What is the server's total sending rate?

(i) 60 Mbps

(ii) Without a server, each laptop sends its stream to each of the other four. How much does each laptop upload?

(ii) 12 Mbps

ANSWER

(i) $5 \text{ streams} \times 4 \text{ copies} \times 3 \text{ Mbps} = 60 \text{ Mbps}$

(ii) $4 \text{ copies} \times 3 \text{ Mbps} = 12 \text{ Mbps}$

Through a server, $N(N - 1)$ streams leave it; in a mesh, $N - 1$ per laptop.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

How many round trips would DNS over TCP take?

SHORT QUESTION · DNS

3. A DNS lookup is one query and one reply.

(i) Which transport protocol does DNS normally use?

(i) UDP

(ii) If DNS used TCP instead, and the TCP handshake takes 1 RTT, how many RTTs pass before the client has the reply?

(ii) 2 RTTs

ANSWER

(i) UDP: one datagram each way, no connection to set up

(ii) 2 RTTs: 1 for the handshake, 1 for the query and its reply

UDP sends the query at once; TCP first spends one RTT on the handshake.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

Which DNS problem does copying not fix?

SHORT ANSWER · DNS

4. Keeping all DNS data on one central server has four problems. Copying the whole database onto many servers fixes three of them: query load, a single point of failure, and long distances to clients. What is the fourth problem, and how does DNS fix it?
-

ANSWER

Who may write: every copy is still edited by one owner.

The fix: split the names into zones along the dots.

Each owner edits its own zone, and each zone is then copied.

Copying fixes load, failure and distance; zones fix who may edit.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

When does a cached record stop being used?

SHORT QUESTION · DNS

5. A resolver received the address record for shop.example.com at $t = 0$, with a TTL of 300 s. Is a query for shop.example.com answered straight from the resolver's cache

(i) at $t = 200$ s?

(i) Circle one: ☒ yes ☐ no

(ii) at $t = 400$ s?

(ii) Circle one: ☐ yes ☒ no

ANSWER

(i) yes: at 200 s the record is still within its TTL

(ii) no: it expired at $t = 300$ s, so the resolver asks again

A record is used until its TTL runs out; then the resolver asks again.

1 The paper

2 Chapter 1

3 Chapter 2

4 Chapter 3

5 Practice

How should you prepare?



Redo the numbers

every numeric example
on the slides,
with a calculator



Draw the exchanges

each protocol as a
sequence diagram,
one event at a time



Explain each point

every knowledge point
in this deck, aloud,
in your own words



Watch the clock

80 minutes for
100 points: under
a minute per point

Work the examples, draw the exchanges, and explain every point here.